

Temporal Training Data

A Literature Review of Methods, Challenges, and Applications

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Abstract

This paper is a literature review of temporal models, a class of AI systems designed to learn from time-series data. We synthesise findings on the main characteristics of temporal data, such as trends, seasonality, stationarity, and noise. We compare classical approaches (e.g., ARIMA) with more recent neural methods, including current architectures and time-aware transformers. The review identifies recurring challenges in the literature, including data quality and irregularity and concept drift. We also examine applications and emerging research directions, such as explainability, and federated and continual learning. By consolidating diverse strands of research, this review clarifies the current state of the field and points to opportunities for future work on temporal AI systems.

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1 Introduction

Artificial intelligence (AI) systems increasingly rely on real-time and evolving data. Models designed to process this data are commonly referred to as temporal models [6]. These use the sequential nature of the data to make predictions and detect patterns over time. Temporal data can take many forms, from sensor readings and financial transactions to user interactions and clinical measurements. Unlike static datasets, temporal data captures not only what happens, but also when it happens.

The increasing availability of temporal data in fields such as finance, transportation, and recommender systems has resulted in advancements in temporal modelling techniques. These models allow for the prediction of future occurrences, the detection of anomalies, and the adaptability to changing settings. However, the very qualities that make temporal data powerful (ordering, unpredictability, changing patterns) also have distinct obstacles.

Temporal models must remain robust in the face of incomplete or irregular data, random noise, evolving patterns (concept drift), and the risk of reinforcing biases. Furthermore, evaluation and benchmarking are more complex for temporal systems, as standard cross-validation approaches often violate temporal causality.

This paper is structured as a literature review. We synthesise prior work on temporal data and modelling, with a focus on identifying recurring themes, comparing classical and neural approaches, and highlighting open research challenges. This review was conducted by surveying recent literature from 2010–2025, with an emphasis on work published in journals, AI conferences, and preprint archives. Sources were selected using the following criteria, (i) proposed methodological innovations, (ii) addressed practical challenges in real-world temporal data, or (iii) provided comprehensive surveys. Earlier foundational works were included to introduce long-standing concepts. This selection strategy aimed to cover breadth and focus on current debates and open problems in temporal modelling.

2 Background

To get a better understanding of the current challenges and applications of temporal models, it is important to understand the characteristics and types of temporal data first.

In the literature there are multiple terms often used interchangeably, or describing something slightly different. Temporal data is a broad term used to describe any data indexed by time. Time-series data refers to sequences of observations collected at successive, usually uniform, time intervals. Temporal data is often used in the broader computer science and database context, while time-series data is more common in statistics and

forecasting.

Temporal models can be neural or statistical models designed to work with temporal data. If these models aim to predict future values on historical observations, they are usually referred to as time-series forecasting models. Temporal data can also be spatio-temporal data, where the data is not only time indexed but also spatially indexed. While the distinctions between these terms are useful for precision, they are not always strictly enforced in practice.

This paper uses the term temporal data, which includes time-indexed data of various forms, and temporal models, which refers to modelling approaches (classical or neural) that operate on such data.

2.1 Temporal Characteristics

Temporal data comes with specific characteristics, which need to be introduced briefly. These characteristics directly influence the analysis of temporal data.

Trend

A trend refers to a systematic, long-term change in the mean of the data over time. In Figure 1, the steady upward shift in U.S. gas consumption illustrates such a trend, where the underlying average increases across decades. Many real-world datasets have changing patterns with a changing mean. These changing mean structures are called trends [12].

Trends can be deterministic or stochastic. A deterministic trend is a predictable and smooth function, for example a linear or quadratic function. This kind of trend reflects a consistent direction of change and is assumed to apply indefinitely, which makes it useful but also potentially risky if applied too generally. A stochastic trend is the cumulative effect of random fluctuations.

Stationarity

Stationarity describes the property that a process remains consistent over time. Stationarity can be strict or weak. Strict stationarity describes a process whose joint probability distributions do not change over time. The process behaves the same as it did in the past and will in the future.

Weak stationarity is used more in practice. It only requires the mean to be constant. For many applications in AI, especially when data is assumed to be Gaussian, this weak form is sufficient. [12].

Seasonality

Temporal data is seasonal if it follows a recurring pattern over fixed time intervals. In Figure 1, the recurring peaks and troughs across years correspond to annual seasonal effects. Unlike trends, which describe long-term directional change, seasonality captures systematic fluctuations that repeat with a known period (e.g. monthly, quarterly, or annually) [35]. Similar to seasonality are cyclic patterns, these are seasonal patterns that do not follow a fixed interval. Modelling cyclic patterns can be harder because of their variable interval length.

Noise

When collecting real-world data, a perfect measurement is rarely achievable and systems are influenced by many unpredictable factors. Noise reflects short-term irregular fluctuations not explained by trend or seasonality. Figure 1 shows the noise in the provided data in the last graph. Temporal models need to account for these characteristics to avoid forecasting errors or wrong patterns being learned. It is important to incorporate robust filtering and smoothing to handle the noise and limit the negative effects [13].

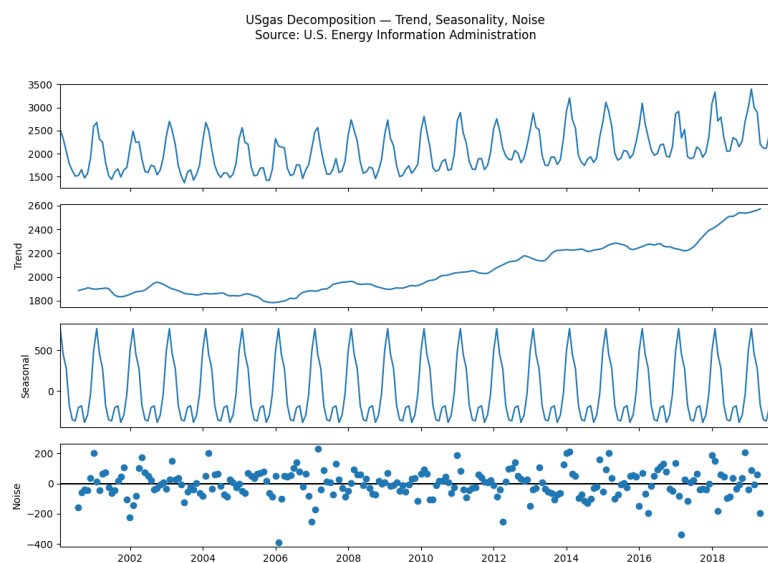


Figure 1: Author's own illustration, based on U.S. Energy Information Administration data (2000–2020)

2.2 Types of Temporal Data

While these characteristics define how temporal data behaves over time, the structure and form of temporal data also plays an important role. This section outlines the key types of temporal data, which are dimensionality, sampling regularity and data availability.

Dimensionality

Time series data can be univariate or multivariate. Univariate data are observations of a single variable recorded over time. Figure 2 illustrates this with JFK airport temperature records, showing a single dimension evolving over time. These series are simpler to analyse and model, but have less context than multivariate data. Multivariate data consists of multiple related variables [22]. As seen in Figure 3, temperature, humidity, and wind speed evolve jointly, requiring models that capture cross-variable dependencies. Multivariate data allow more complex modelling and the ability to capture interactions between variables, but they require more sophisticated methods to handle issues like different scaling, correlations, and missing values.

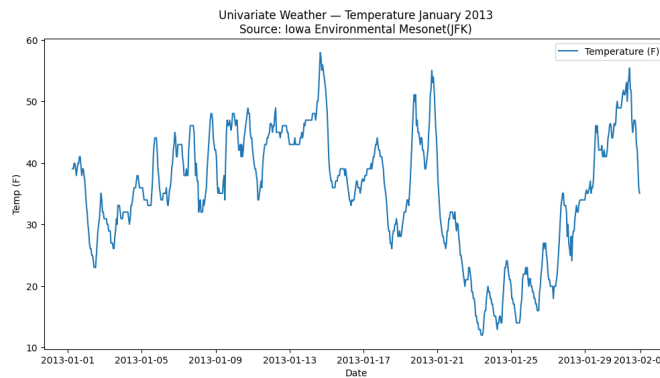


Figure 2: Univariate time-series using weather data from JFK. Author's own illustration.

Sampling regularity

Regularly collected data is being recorded at a predefined and consistent time interval. This simplifies analysis and is common in many statistical models. Irregularly collected data is being recorded with varying time intervals. This data is often based on events rather than a continuous measurement. Figure 4 contrasts regularly sampled intervals

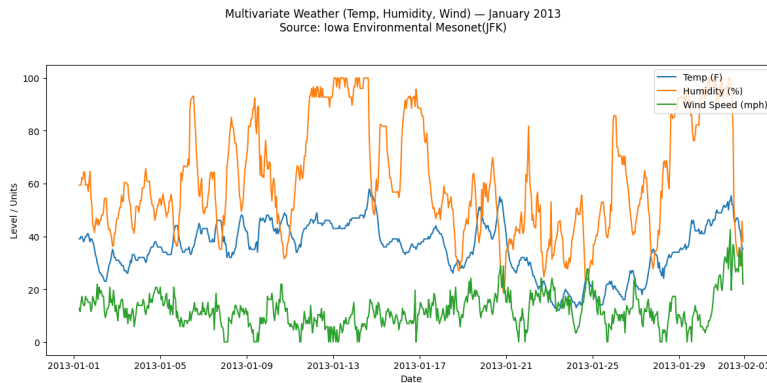


Figure 3: Multivariate (Temperature, Humidity, Wind Speed) time-series using weather data from JFK. Author’s own illustration.

with sporadic ones, emphasizing why irregular sampling poses challenges for classical methods.

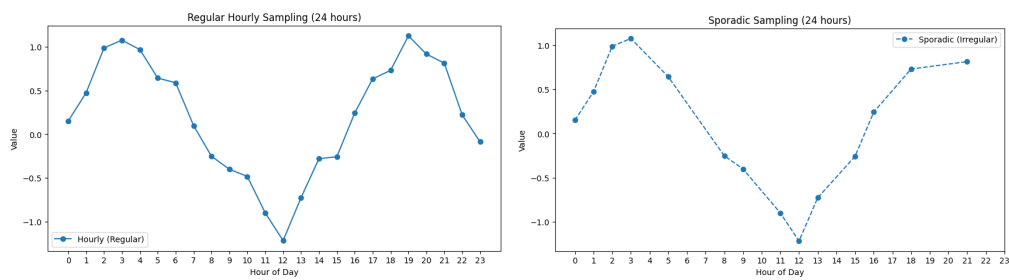


Figure 4: Regular and Sporadic time intervals using synthetic data in a 24-hour interval. Author’s own illustration.

Data availability

Streamed data is often generated continuously in real time and in large volumes. It requires algorithms that can process information incrementally and make predictions in real time, such as Kalman filters. Historical data is static and completely available, no new data will be added. It is suitable for batch-processing approaches like ARIMA or deep learning-based models.

2.3 Modelling Approaches

The following section outlines classical and modern approaches used to capture temporal dependencies. As we already mentioned Kalman filters and ARIMA, we want to introduce these modelling approaches in this section first.

Classical Approaches

ARIMA

A widely used approach is Autoregressive Integrated Moving Average (ARIMA) [11]. ARIMA combines three components: an autoregressive (AR) part that expresses the current value as a linear combination of past values, a differencing (I) step that removes trends to achieve stationarity, and a moving average (MA) part that models dependence on past forecast errors. Formally, a simple ARIMA(p, d, q) process can be written as:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} - \theta_1 \epsilon_{t-1} - \dots - \theta_q \epsilon_{t-q} - \epsilon_t$$

where p is the autoregressive order, d is the degree of differencing applied, q is the moving average order, and ϵ_t is white noise. Here the moving average parameters (θ 's) are defined so that their signs are negative in the equation, following the convention introduced by Box and Jenkins [11].

ARIMA models are well understood, easy to implement, and effective for short-term forecasting of linear and stationary series. They remain a standard baseline in forecasting competitions and industrial applications. ARIMA requires stationarity and struggles with nonlinear dependencies. It is not designed for high-dimensional or irregularly sampled data. Extensions such as SARIMA (Seasonal ARIMA) [23] address seasonal effects, and VARIMA [30] extends ARIMA to multivariate settings, but these approaches still face scalability challenges.

Kalman filters

Another classical approach is the Kalman filter [33], widely used in signal processing and tracking tasks. Kalman filters recursively estimate system states from noisy observations but are limited to linear systems with Gaussian noise. While influential historically, they play a smaller role in current machine learning research compared to ARIMA.

Neural Approaches

Neural networks have become quite dominant for modelling temporal data, as they can capture complex, non-linear dependencies and are able to overcome the limitations of the classical approaches. Major families of neural models include:

1. Recurrent Neural Networks (RNNs)

- RNNs [26] are designed for sequential data by maintaining internal hidden states which change with the input sequence.
- Variants: Long Short-Term Memory (LSTM) [19] and Gated Recurrent Units (GRU) [15].
- Limitations: Struggle with very long sequences (vanishing gradients, scalability).

2. Temporal Convolutional Networks (TCNs)

- Use dilated causal convolutions for sequence modelling [1].
- More parallelizable and stable than RNNs.

3. Transformers

- Transformers rely on self-attention, which computes pairwise relationships between all time steps in a sequence [27].
- Captures both short- and long-range dependencies without recursion.
- Positional encodings assume regular sampling.
- Variants: Time Series Transformers and Informer [37]

4. Time-aware Transformers

- Addresses irregular sampling and time gaps.
- ChronoFormer integrates continuous-time embeddings and hierarchical attention to capture irregular time gaps and temporal patterns [36].
- Zhang et al. introduced FTL-Trans, a time-aware transformer-based model for clinical outcome prediction, and demonstrated that it outperforms other transformer-based architectures [34].

3 Current Challenges

While these approaches offer powerful ways to work with temporal data, real-world scenarios present a set of persistent challenges that complicate their usage.

3.1 Data Quality and Complexity

Temporal datasets in real-world applications often show issues of incompleteness, noise, and irregular sampling intervals [17]. Sensors may drop readings or record at inconsistent time steps, leading to missing values and uneven time gaps in the data. Such irregularities complicate modelling. Noise in time-series data may hide true patterns and provide false correlations. In contrast to static datasets with fixed labels, in a time-evolving context, the ground truth can change over time. Therefore, Temporal data needs dynamic labelling strategies. Mislabelling or inconsistent annotation over time can introduce bias. Specialised methods are necessary to preserve temporal label consistency and quality in reference data, as errors or biases in time-dependent labels could compromise model reliability as they scale over time.

3.2 Concept Drift

Concept drift is a fundamental challenge with temporal data, where the data-generating distribution changes [18]. This violates the assumption that training data and future data come from the same distribution. When concept drift occurs, the learned patterns do not match the new data. The model then degrades without retraining, as the new data does not resemble the training data. To avoid concept drift, it is necessary to implement a detection and adaptation mechanism [16]. The incoming data needs to be monitored for any signs of drift. Common adaptation algorithms are incremental learning [29] or periodic retraining [32]. A combination of automated drift detection and human-in-the-loop¹ is often used in practice, especially in critical systems [2].

3.3 Bias, Fairness, and Ethics

Bias, Fairness and Ethics need to be discussed for every AI implementation. Temporal Data amplifies bias and fairness over time, therefore these need to be actively managed. Even initially fair models have been observed to drift as they age and process new data [28]. This occurs due to changes in the significance or distribution of features associated with groups, or because the model begins to adapt to anomalies in new data that have sensitive qualities. This risk of unintended discrimination marks the importance for continuous fairness and bias monitoring. Periodical re-evaluation of fairness metrics is often used to circumvent a shift in fairness [3]. AI systems that make sequential decisions or operate in a feedback loop can affect future data and results, which can reinforce bias [25].

¹Human-in-the-loop is a system design model that integrates human expertise and oversight into automated processes. In this context, a human is used to oversee the drift detection.

Polymakers and practitioners need to develop fairness criteria that stay fair long-term. Incorporating domain knowledge, monitoring, and limiting the autonomy of models might be necessary to prevent negative long-term consequences.

3.4 Evaluation and Benchmarking

Standard evaluation techniques are not suitable for temporal systems, because they assume data points are independent and identically distributed. Training sets should not include information from after the test set when dealing with temporal data as this would violate causality. Rolling origin (forward chaining) cross-validation works by training the model on an initial window of time, then testing on the next time period, then retraining by extending the window [8]. This mimics real forecasting and respects the time order for each window. Another technique is to split the data with a hold-out period at the end to ensure the test set is always chronologically after the training set. Both methods ensure that the evaluation metrics reflect the actual model's performance. Building on these evaluation techniques, temporal benchmarks provide standardised datasets for consistent, time-aware evaluation.

There are a lot of benchmarks for non-temporal systems, but temporal systems tend to use proprietary datasets, which makes it difficult to replicate results [7]. It becomes challenging to identify which approaches are state-of-the-art or to quantify improvements objectively. Initiatives like the NeurIPS competitions on data streams provide shared resources to measure and compare the competitors. Establishing common benchmark datasets and metrics for temporal learning remains an important challenge. Recently, Kraus et al. showed that pretraining without labels on diverse temporal datasets is possible, thereby challenging that multi-dataset pretraining is ineffective for time-series data and mitigating the need for expensive labelled datasets [21].

4 Current Applications

Despite the current challenges, temporal models are used in many domains. This section shows how temporal models are currently applied and future research directions.

4.1 Real-World Applications

Finance

In the finance sector, temporal models are used to analyse streams of stock prices and economic indicators to make trading decisions in real time. Financial data is notoriously

volatile and non-stationary. It is characterized by noisy, irregular fluctuations and sudden shifts, therefore models need to handle abrupt changes in patterns [24]. Forecasting processes such as volatility prediction are crucial for tasks such as derivative pricing and risk assessment, but effectively anticipating market behaviour is challenging due to the inherent complexities involved [14].

Mobility and Transportation

Uber leverages forecasting models to predict rider demand and driver supply across locations in advance [5]. These models direct drivers to areas where rides are likely to be requested, which improves efficiency and reduces wait times. The models used by Uber introduce additional complexity because they use spatio-temporal data. As Uber's engineers note [5], operating in the physical world means forecasts must account for road network capacity and geographic distance. Uber has the advantage of a very large proprietary data set to train a specialized model. This dataset shows strong weekly patterns and growth trends.

Recommender Systems

A recommender system is used to recommend content to a user, often employed in social media, streaming and shopping websites. Informally, a recommender system is often referred to as "the algorithm", which aims to show the user content which is similar to the content consumed and potentially interesting to the user. Modern recommender systems increasingly incorporate temporal dynamics to improve recommendations as user behaviour evolves. Traditional collaborative filtering algorithms assumed stationarity in user-item interaction data, failing to account for a drift in user preference [31]. Temporal recommender systems track the order of user interactions and captures temporal patterns in consumption. Platforms like Netflix use session-based and temporal recommendation models that adjust to trends [10]. Recommender systems are prone to drift and have difficulties handling data sparsity. A recommender system needs to continuously learn from new interaction data as old data fades. The use of time as a context has proven important for avoiding outdated content and personalising recommendations based on the current context.

4.2 Emerging Research Directions

Beyond these established use cases, research continues for temporal models.

Explainability in temporal models

Explainability aims to explain what has been done, what is being done, and what will be done next, and to unveil which information these actions are based on. Kraus et al. (2024) propose the “Right on Time (RioT)” approach, which interacts with model explanations in both the time domain and frequency domain to guide the learning process [20]. By using feedback on which signal properties are significant, the model can be forced to rely on annotated false correlations. This dual-domain explanatory feedback has been demonstrated to improve the reliability of time series classifiers. The goal is to ensure that temporal models not only predict properly but also explain why a prediction was made, which is a key feature in establishing confidence.

Federated/continual learning

Real-world temporal data often arrives as a continuous stream and may be distributed across multiple origins. Federated learning (FL) and continual learning (CL) approaches can also use streams as their data source. Federated learning enables training a global model on decentralized data sources without requiring the raw data to be centralized. FL can be used for sensitive temporal data like health records since it preserves privacy by design [4]. Researchers are developing FL frameworks for traffic forecasting and disease detection, showing that models can be trained collaboratively across multiple data origins.

Continual learning addresses the issue of concept drift. Instead of training once on a fixed dataset, continual learning algorithms update the model incrementally as new data arrives, while avoiding losing past knowledge [9]. Continual learning works by periodically retraining on new data, replaying old data or expanding the models’ capacity for new patterns.

5 Conclusion

Temporal models are widely used in domains where decisions depend on time-sensitive information. This review has shown how advancements in neural architectures have improved the ability to capture temporal dependencies and be more robust to concept drift, irregular sampling and fairness concerns.

Addressing this tension between accuracy and accountability requires not only technical innovation but also shared benchmarks, transparent evaluation protocols, and domain-specific fairness criteria.

New research into time-aware transformers, federated and continual learning offers



promising potential. However, they come with their own set of challenges which need to be overcome.

Future progress will depend on tighter integration between theory, practice, and policy. Interdisciplinary collaboration will be important to ensure that temporal AI systems are trustworthy, ethical, and resilient in dynamic settings. By systematically addressing these challenges, the field can move beyond short-term performance toward temporal modelling that supports long-term benefit.

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